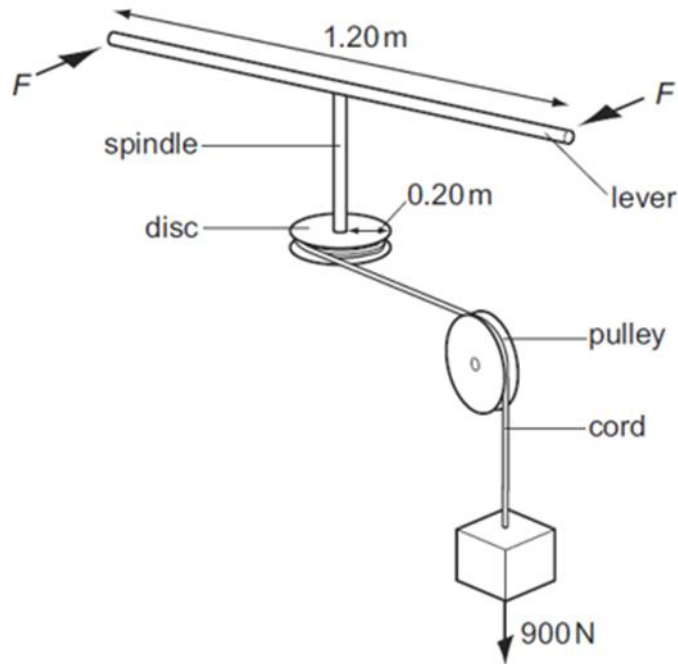


7. A spindle is attached at one end to the centre of a lever 1.20 m long and at its other end to the centre of a disc of radius 0.20 m. A cord is wrapped round the disc, passes over a pulley and is attached to a 900 N weight.



What is the minimum force F , applied to each end of the lever, that could lift the weight?

- A 75 N **B 150 N** C 300 N D 950 N

Answer: B

torque couple on lever = $F \times 1.20$

moment on disc = 900×0.20

To lift weight, $F \times 1.20 > 900 \times 0.20 \Rightarrow F > 150 \text{ N}$